Research Design: Home Run Derby Participation and Change in Batted-Ball Approach

This design builds on the domain and skill review by narrowing the research question from performance to approach. Performance statistics such as on-base plus slugging (OPS) and wins above replacement (WAR) reflect pitching, defense, and luck in addition to a hitter's own swing, so a true change in approach can be masked or exaggerated over a second half of roughly 65 to 70 games. Batted-ball data describes a hitter's approach more directly. The research question: how, if at all, does a Home Run Derby participant's batted-ball profile change from the first half to the second half of the season, relative to a matched group of non-participating All-Stars?

The cause under study is Derby participation; the proposed effect is a shift toward a more pull-side, air-ball-dominant profile, since pulling the ball in the air is the easiest path to a home run (the wall is shallower down the lines). That requires contact earlier in the pitch's flight, and Giordano et al. (2026) tied contact depth to swing attack angle, while Nakashima et al. (2025) showed the acceptable timing window at impact is only several milliseconds and depends on swing path. Repeated maximal-effort, pull-air swings could plausibly leave a trace in a player's regular swing timing, which this design tests for directly rather than assuming.

Purpose and Causal Logic

This is a non-experimental study. A player is invited based on first-half home run totals and can accept or decline the invite, so participation is self-selected on exactly the kind of hitter who already favors an extreme pull-air approach. The design can at most show whether participants' profiles shifted more than a matched comparison group's, after accounting for first-half level to net out regression to the mean (Barnett, van der Pols, & Dobson, 2005). It cannot rule out unmeasured confounding variables (e.g. an unrelated coaching change or a significant

adjustment in personal life) and cannot generalize beyond the seasons and formats studied. It supports a claim of association under a controlled comparison, not causation.

The comparison group pairs each participant with a non-participating All-Star of similar first-half batted-ball approach, matched 1:1 by Mahalanobis distance on the standardized variables defined below (Stuart, 2010) rather than by a shared cluster label, following the spirit of DeFroda et al. (2021) and Sobol et al. (2024). In a scenario in which no non-participant falls within a pre-set maximum matching distance, the design falls back to a within-subject comparison following Jaiclin and McCollum (2010): each participant's Derby-year shift against their own shift in seasons without participation, which removes between-player confounds but requires multiple seasons per player and reintroduces era-level differences (such as the change in Home Run Derby format in 2026) as a confound of its own.

Data

Data comes from three public sources, summarized in Table 1 and cited in the reference list: Baseball Savant (n.d.), FanGraphs (n.d.), and Baseball-Reference (n.d.), all used routinely in peer-reviewed baseball analytics work (Koseler & Stephan, 2017). Savant's data is available for public research and analysis; FanGraphs and Baseball-Reference permit non-commercial academic use with attribution. No data will be resold or redistributed.

One variable is hand-coded rather than pulled directly: home-run-specific spray direction, split by half-season, since Savant exports overall batted-ball direction only as a season aggregate. Each home run is classified as pull, middle, or opposite field from the player's Statcast spray chart (handedness defines "pull") using a codebook fixed before coding begins.

Table 1

Data Sources and Use Policies

Source	Primary variables pulled	Use policy
Baseball Savant	Exit velocity, launch angle, pull/oppo%, barrel%, bat speed/attack angle (2023+), HR spray charts	Public research/commentary use
FanGraphs	Split-season batted-ball direction/trajectory (Pull%, Cent%, Oppo%, GB%/FB%/LD%) via the Splits Leaderboard's 1st/2nd Half preset; plate appearances	Non-commercial academic use, attribution
Baseball-Reference	Season schedule, All-Star break date, All-Star rosters, split stats	Non-commercial academic use, attribution

The three sources use three different player identifiers, merged via a crosswalk (the Chadwick Bureau register, through pybaseball) rather than by name, since names collide and are spelled inconsistently across sites. The data form a player-season-half panel, with multiple seasons pooled once collection is complete. Inclusion criteria are a minimum plate-appearance total in both halves and position players only. Timer-based seasons (2015-2025) and the swing-based 2026 format are kept as separate regimes, and bat-tracking variables (bat speed, attack angle) are restricted to 2023 onward due to not becoming public, polished data until that time. All data is public, on-field performance with no institutional review concerns. The main risks are the manual coding, addressed above, and match availability, addressed by the matching and backup-design logic above.

Statistical Analysis

Before comparison, batted-ball and swing-trait variables are standardized to z-scores using the first-half mean and standard deviation of the full eligible sample. These variables include Pull% and FB% from FanGraphs' Batted Ball splits, average launch angle, average exit velocity, strikeout rate, HR Pull% and HR/FB%, and, where available, bat speed and attack angle. Second-half values for the same players use those same first-half reference statistics rather

than being re-centered, so a change in standardized score reflects a real shift in level rather than an artifact of the comparison. Initially this design planned to reduce these variables via PCA before clustering hitters into approach groups; that step is dropped, since a component's loadings are not guaranteed to track something recognizable as "pull-air approach" rather than some other mix the data happen to produce (Jolliffe & Cadima, 2016), and using the named variables directly keeps every result traceable to a specific, interpretable measurement.

Hitters are matched directly on these standardized first-half variables rather than grouped into clusters, since a shared cluster can still mask real differences between hitters that matter for this comparison. Each participant is paired with the closest available non-participant by Mahalanobis distance (Stuart, 2010); a pre-set maximum matching distance keeps a poorly matched participant from being forced into a pair, sending them to the within-subject backup design instead, and match quality is confirmed via standardized mean differences between matched pairs (Stuart, 2010).

One outcome is designated the primary, confirmatory test ahead of time and tested once, not re-tested under several specifications. The ideal measure would be the share of batted balls that are both pulled and hit in the air, but that joint rate is not available without re-deriving spray-angle buckets from raw Statcast coordinates ourselves, a step avoided since Statcast's exact classification is not published and cannot be independently verified. Instead, the primary outcome is a Pull-Air Index: the average of a player's standardized Pull% and standardized FB%, both from FanGraphs' Batted Ball splits for the same half, keeping both ingredients in one classification system. The remaining standardized variables of HR Pull% and HR/FB% more directly capture pulled contact in the air on a much smaller per-half sample of home runs only. These are treated as secondary, exploratory comparisons: informative, but not used alone to

claim a confirmed effect. This limits the chance that testing many outcomes turns up an isolated significant finding by chance alone.

For the primary outcome, a paired t-test tests whether each participant's first-half and second-half Pull-Air Index differ, and an independent-samples t-test compares the size of that change against matched controls, reported as a 95% confidence interval rather than a p-value alone given a sample only slightly above 100. Secondary outcomes use the same comparisons with a Bonferroni-corrected threshold (alpha = .05 divided by the number of secondary tests) to limit the false-positive rate from running several tests (Armstrong, 2014), and are reported as exploratory.

Because timer-based (2015-2025) and swing-based (2026) formats are kept as separate regimes, the design also checks whether the Index shift differs across the four groups formed by crossing regime with participant status, using a one-way ANOVA followed by Tukey's Honestly Significant Difference (HSD) test for the pairwise contrasts rather than six unprotected comparisons (Abdi & Williams, 2010); this regime check is exploratory, reported alongside the primary test rather than as a second confirmatory claim.

Last, a multiple regression predicts each participant's second-half Pull-Air Index from Derby participation, first-half Index (netting out regression to the mean; Barnett et al., 2005; Jaiclin & McCollum, 2010), and format regime, mirroring DeFroda et al.'s (2021) multivariable approach applied to approach instead of performance. Regression assumptions, matching balance, and priori power calculation will be checked before interpretation, so a null result is judged against a realistic detectable effect size rather than assumed to mean no effect exists.

References

Abdi, H., & Williams, L. J. (2010). Tukey's honestly significant difference (HSD) test. In N. J. Salkind (Ed.), Encyclopedia of research design (pp. 1–5). SAGE Publications. https://doi.org/10.4135/9781412961288.n478

Armstrong, R. A. (2014). When to use the Bonferroni correction. Ophthalmic and Physiological Optics, 34(5), 502–508. https://doi.org/10.1111/opo.12131

Barnett, A. G., van der Pols, J. C., & Dobson, A. J. (2005). Regression to the mean: What it is and how to deal with it. International Journal of Epidemiology, 34(1), 215–220. https://doi.org/10.1093/ije/dyh299

Baseball-Reference. (n.d.). Baseball-Reference.com [Data set]. Retrieved October 2026, from https://www.baseball-reference.com/

Baseball Savant. (n.d.). Statcast search [Data set]. Retrieved October 2026, from https://baseballsavant.mlb.com/

DeFroda, S. F., Staffa, S. J., Keeley, T., & Kriz, P. K. (2021). Home Run Derby participation in Major League Baseball players: Is there associated injury risk and impact on second-half performance? Orthopaedic Journal of Sports Medicine, 9(2), 1–8. https://doi.org/10.1177/2325967120983350

FanGraphs. (n.d.). Splits leaderboards [Data set]. Retrieved October 2026, from https://www.fangraphs.com/

Giordano, K. A., Jump, I., Lerch, B., & Oliver, G. D. (2026). Contact depth affects most of the other hitting metrics we analyze. Sports, 14(8), Article 330. https://doi.org/10.3390/sports14080330

Jaiclin, M., & McCollum, J. (2010). Home Run Derby curse: Fact or fiction? Baseball Research Journal, 39(2). https://sabr.org/journal/article/home-run-derby-curse-fact-or-fiction/

Jolliffe, I. T., & Cadima, J. (2016). Principal component analysis: A review and recent developments. Philosophical Transactions of the Royal Society A, 374(2065), Article 20150202. https://doi.org/10.1098/rsta.2015.0202

Koseler, K., & Stephan, M. (2017). Machine learning applications in baseball: A systematic literature review. Applied Artificial Intelligence, 31(9–10), 745–763. https://doi.org/10.1080/08839514.2018.1442991

Nakashima, H., Horiuchi, G., Kimura, A., & Sakurai, S. (2025). Acceptable range of timing error at bat-ball impact in baseball depends on the bat swing path. Frontiers in Sports and Active Living, 7, Article 1557145. https://doi.org/10.3389/fspor.2025.1557145

Sobol, G. L., Rosenberg, S. I., Baker, H. P., Dayton, S. R., Plantz, M. A., & Tjong, V. K. (2024). National Basketball Association slam dunk contest participation does not lead to increased injury risk or adversely affect player performance. Current Orthopaedic Practice, 35(6), 255–259. https://doi.org/10.1097/BCO.0000000000001278

Stuart, E. A. (2010). Matching methods for causal inference: A review and a look forward. Statistical Science, 25(1), 1–21. https://doi.org/10.1214/09-STS313